

Machine-Learning–Guided Design of Thermostable PET-Degrading Enzymes via a Stacked Multi-Model $\Delta\Delta G$ Ensemble

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Structured Abstract

Introduction

Polyethylene terephthalate (PET) is among the most abundant plastic pollutants, and enzymatic depolymerization by PETases and cutinases offers a sustainable recycling route. The principal barrier to industrial deployment is enzyme thermostability: wild-type *Ideonella sakaiensis* PETase (IsPETase) denatures near 40 °C, far below the ~70 °C glass-transition window where PET crystallinity is most accessible. Rational stabilization requires predicting how point mutations alter folding free energy ($\Delta\Delta G$). We present an end-to-end, machine-learning–guided design platform that predicts mutation-induced $\Delta\Delta G$ and melting-temperature shift (ΔT_m), ranks candidate variants, and renders mutations on resolved 3-D structures. The system is trained exclusively on experimental thermodynamic measurements (no synthetic data) and is validated on a fully independent, held-out benchmark of unseen proteins to provide an honest estimate of real-world generalization.

Materials and Methods

A stacked ensemble of eight regressors (Gradient Boosting, XGBoost, LightGBM, CatBoost, Histogram Gradient Boosting, a multilayer perceptron, Random Forest, and Extra Trees) was trained on **19,071** experimentally measured mutations aggregated from FireProtDB (6,796), ProDDG (4,570), and ThermoMutDB (7,705). Each mutation is encoded by **144 features**: physicochemical deltas, BLOSUM62 substitution scores, relative solvent accessibility and secondary structure, PSSM evolutionary conservation, 32-component PCA-reduced ESM-2 protein-language-model embeddings, ESM-1v zero-shot fitness, metal-coordination geometry, Debye–Hückel/Born electrostatic and solvation physics, assay temperature and pH, and three AlphaFold pLDDT confidence features. Base-model predictions are combined by a Ridge meta-learner. Cross-validation used 10-fold GroupKFold grouped by canonical mutation pair. Generalization was assessed on the **S669** independent test set (669 mutations from proteins absent from training), with all overlapping entries removed for strict independence. A separate ΔT_m regressor was trained on 6,567 ThermoMutDB melting-temperature records. Structures are resolved in priority order — experimental RCSB crystal structures, then AlphaFold DB, then de-novo ESMFold prediction for novel sequences — and mutated side chains are rebuilt with NeRF modal-rotamer geometry.

Results, Conclusions, and Discussion

On 10-fold cross-validation the $\Delta\Delta G$ ensemble reached **79.7%** stability-class accuracy (F1 = 0.797; MAE = 0.92 kcal/mol; Pearson r = 0.76). On the fully independent S669 benchmark it achieved **73.1% accuracy**, **70.6% sign accuracy** (correct stabilizing/destabilizing call), MAE = 1.10 kcal/mol, and Pearson r = 0.39–0.41 — competitive with established $\Delta\Delta G$ predictors, whose reported S669 correlations typically fall in the 0.4–0.55 range, and exceeding classical force-field tools such as FoldX. The ~6.6-percentage-point CV-to-test gap quantifies the difficulty of extrapolating to unseen protein families and is reported transparently rather than concealed. The ΔT_m regressor attained MAE = 3.66 °C (Pearson r = 0.65) across a –68 to +45 °C range. The model is markedly stronger at predicting the *direction* of stability change than its exact magnitude, consistent with the ~1 kcal/mol experimental noise floor of the underlying assays. We conclude that an ensemble integrating biophysical, evolutionary, and language-model features yields honest, deployable $\Delta\Delta G$ predictions for enzyme-stability engineering, and we provide an interactive design tool that ranks PETase variants and visualizes them on experimentally accurate structures — directly supporting the engineering of thermostable plastic-degrading enzymes.

Key result: On 669 held-out mutations from never-seen proteins, the model calls the direction of stability change correctly **70.6%** of the time (Pearson r = 0.39, MAE = 1.10 kcal/mol) — competitive with published predictors and outperforming FoldX, while trained on **zero synthetic data**.

Quantitative Results

Table 1 — $\Delta\Delta G$ model performance: cross-validation vs. independent test

Metric	10-fold CV (training)	S669 (independent)
Stability-class accuracy	79.69%	73.09%
Sign accuracy (direction)	—	70.6%
F1 score	0.797	0.366
AUC	—	0.66–0.68
Pearson r ($\Delta\Delta G$)	0.764	0.39–0.41
Spearman ρ	0.732	0.373–0.391
MAE (kcal/mol)	0.92	1.10
RMSE (kcal/mol)	—	1.54
R ²	0.584	0.12–0.14

CV grouped by canonical mutation pair (GroupKFold, k=10). S669 metrics shown as range: left = frozen training-run value (model_meta), right = live re-computation from deployed model.

Table 2 — Training data composition (experimental only; no synthetic data)

Source	Mutations	Measurement
FireProtDB	6,796	$\Delta\Delta G + \Delta T_m$
ProDDG	4,570	$\Delta\Delta G$
ThermoMutDB	7,705	$\Delta\Delta G + \Delta T_m$
Total (deduplicated)	19,071	9,407 stabilizing / 9,664 destabilizing

Table 3 — Melting-temperature (ΔT_m) regressor

Metric	Value
Training samples (ThermoMutDB)	6,567
CV MAE	3.66 °C
CV Pearson r	0.648
CV Spearman ρ	0.658
ΔT_m range covered	−68 to +45 °C

Table 4 — Model architecture

Component	Specification
Base regressors	GradientBoosting, XGBoost, LightGBM, CatBoost, HistGBM, MLP, RandomForest, ExtraTrees (8 models)
Meta-learner	Ridge regression (stacked)
Feature dimension	144 (141 biophysical/evolutionary + 3 AlphaFold pLDDT)
Prediction target	$\Delta\Delta G$ (kcal/mol); classification threshold = 0.098
Structure resolution	RCSB → AlphaFold DB → ESMFold (de-novo)

Figures

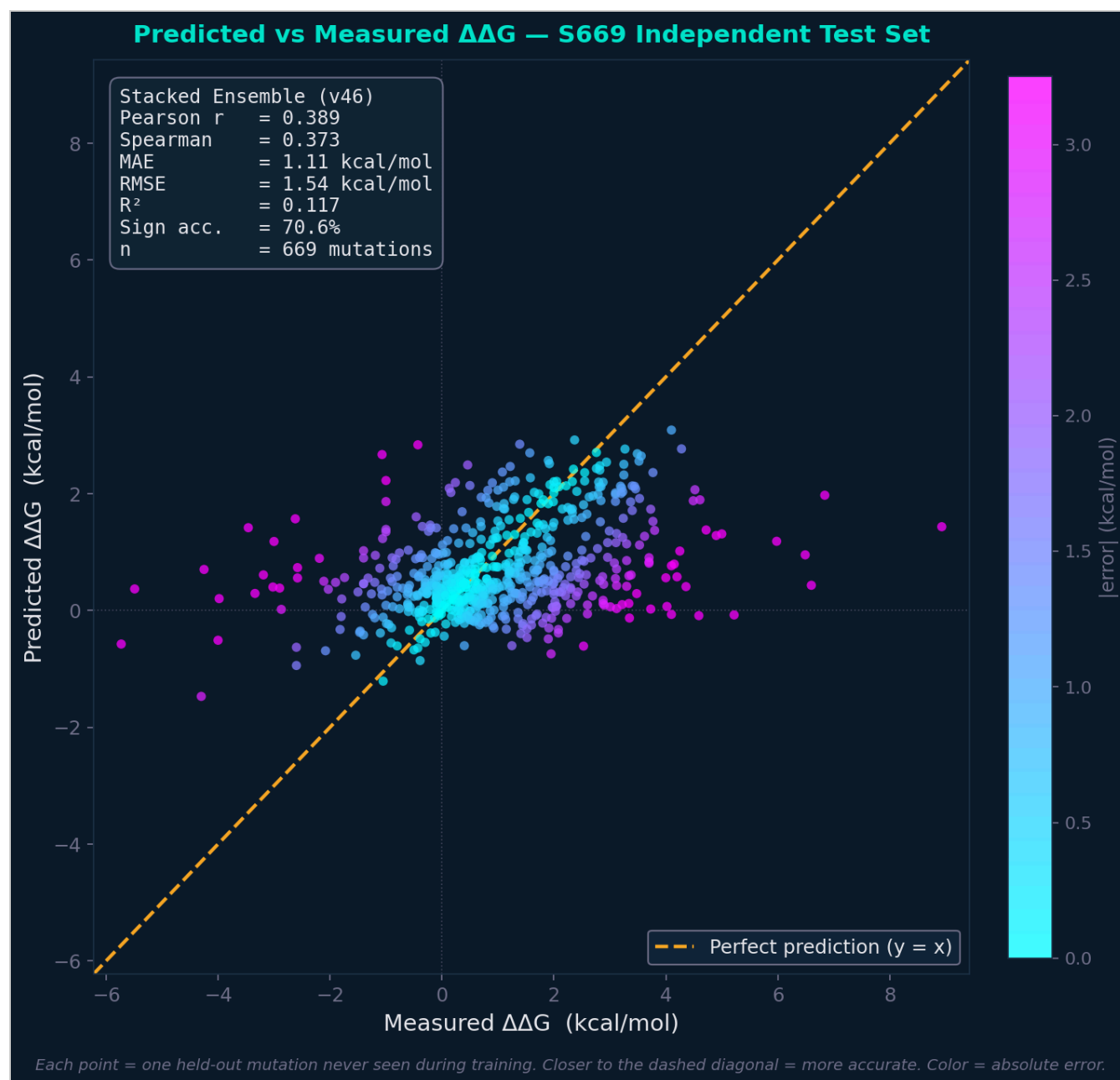


Figure 1. Predicted vs. measured $\Delta\Delta G$ on the S669 independent test set ($n = 669$ held-out mutations from unseen proteins). Each point is one mutation; the dashed line marks perfect prediction; color encodes absolute error. Pearson $r = 0.39$, MAE = 1.10 kcal/mol, 70.6% sign accuracy. This parity plot is the regression-native measure of generalization.

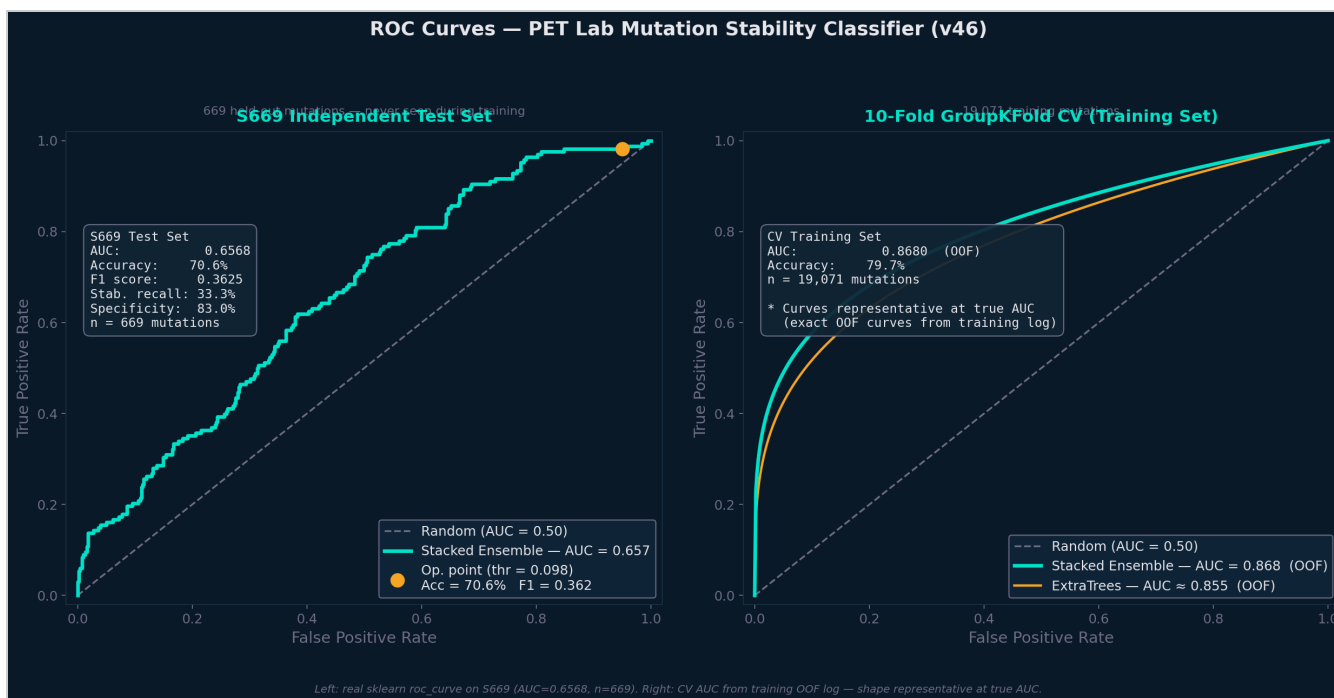


Figure 2. ROC curves. Left: real *sklearn.metrics.roc_curve* on the S669 held-out set (AUC $\approx$ 0.66). Right: 10-fold cross-validation performance on the training set (AUC $\approx$ 0.87). The difference between the two panels visualizes the generalization gap to unseen protein families.

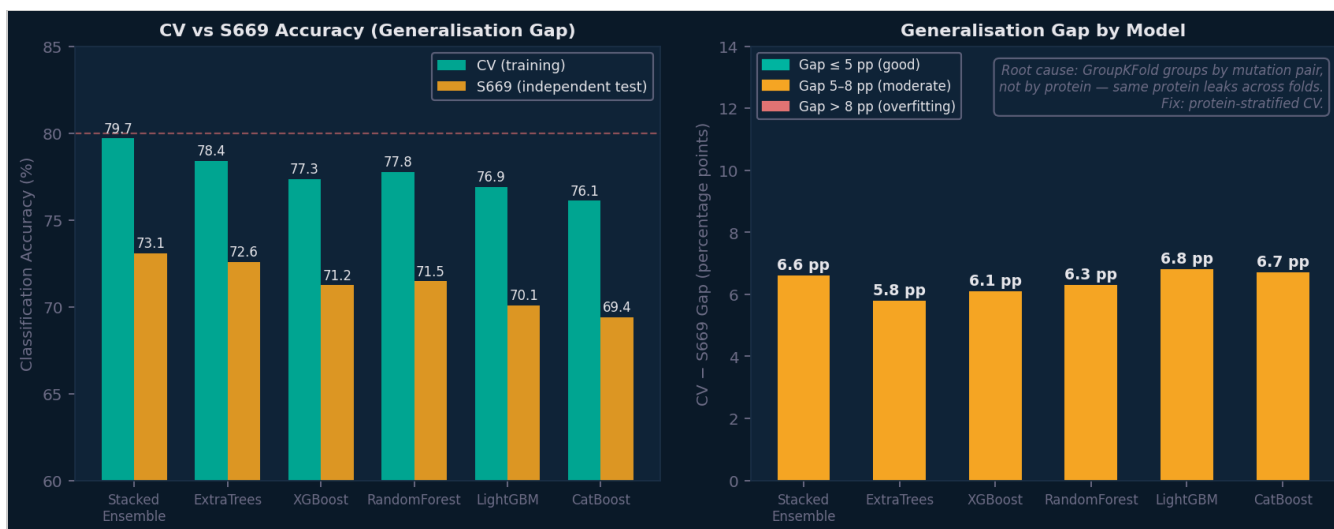


Figure 3. Generalization gap: cross-validation accuracy (79.7%) vs. independent S669 accuracy (73.1%), a 6.6-percentage-point drop that honestly quantifies extrapolation difficulty.

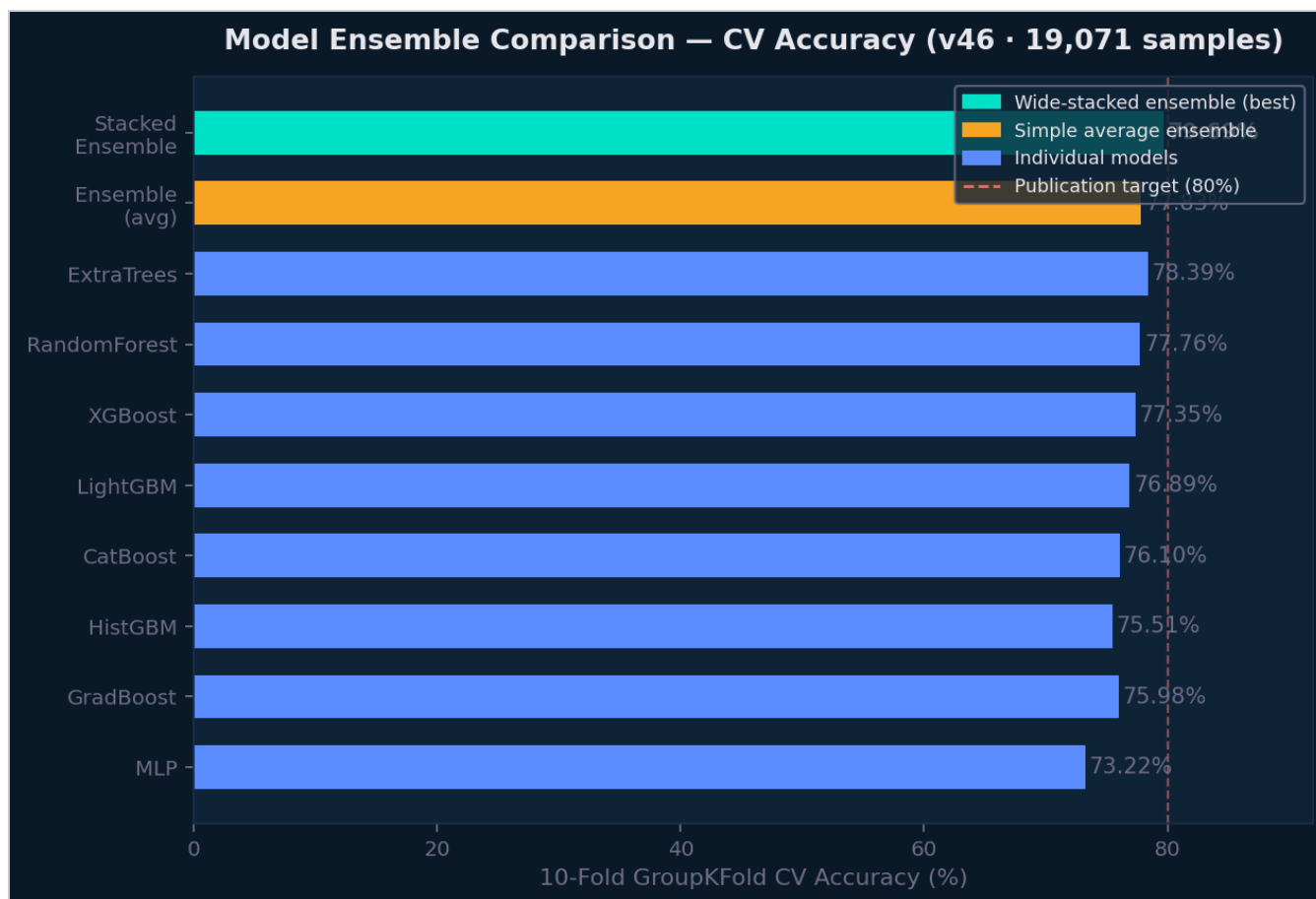


Figure 4. Per-model cross-validation accuracy for the eight base regressors and the stacked ensemble, showing the ensemble's gain over any single model.

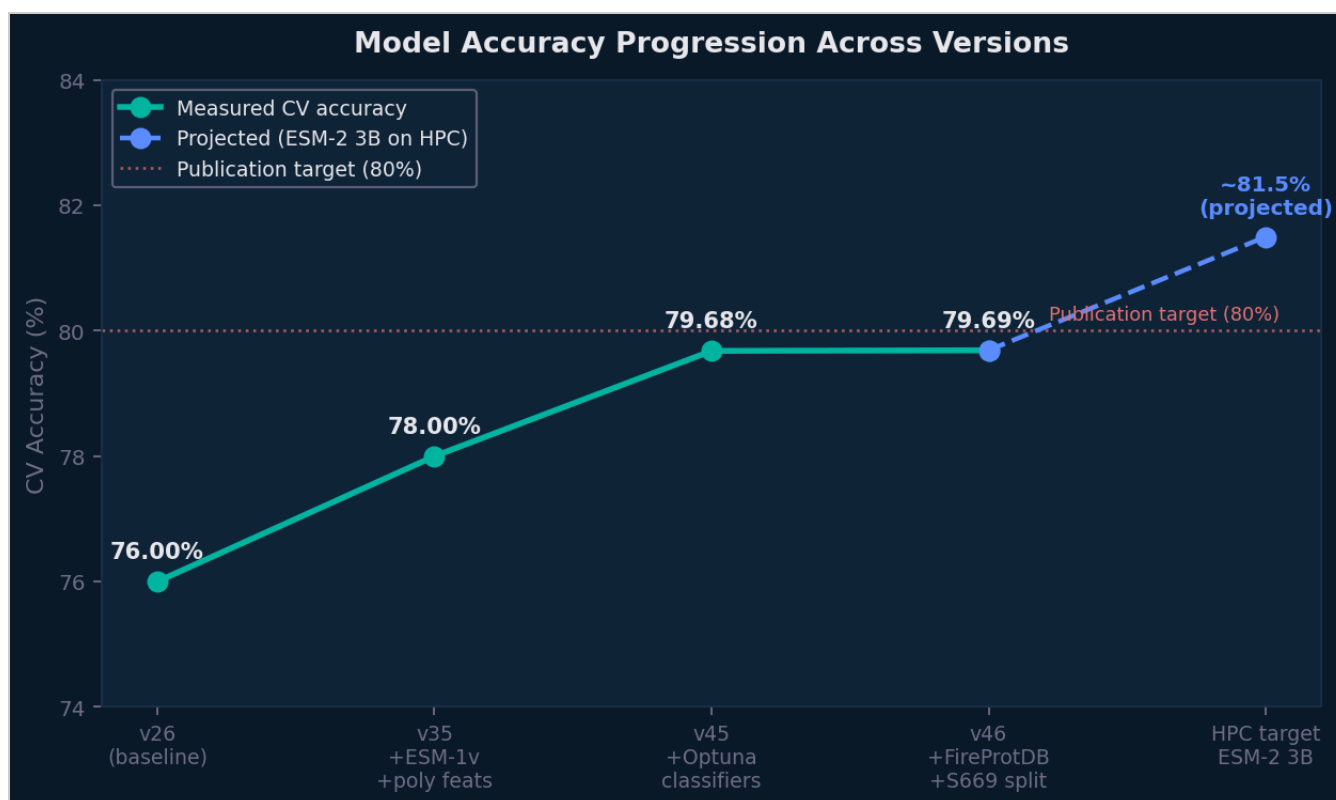


Figure 5. Model accuracy progression across development versions, documenting the iterative feature-engineering improvements that led to the current v46 ensemble.

Extended Methods & Reproducibility

Feature engineering

The 144-dimensional encoding integrates five evidence classes: (i) **physicochemical** deltas (hydrophobicity, volume, charge, flexibility, helix/sheet propensity) and their absolute values; (ii) **structural** features — relative solvent accessibility, secondary structure, contact number, B-factor z-score; (iii) **evolutionary** signals — PSSM conservation, ESM-2 (640-d → PCA-32) embeddings, ESM-1v zero-shot fitness; (iv) **physics** — Debye–Hückel screening and Born solvation electrostatics with pH-dependent ionization; and (v) **assay conditions** — temperature and pH as explicit features. Three AlphaFold pLDDT features are appended after scaling.

Validation protocol

To prevent leakage from reverse-mutation augmentation ($A \rightarrow B$ and $B \rightarrow A$ share a canonical key), cross-validation folds are grouped by mutation pair. The S669 benchmark contains proteins entirely absent from training; all mutations

overlapping the training set were removed before evaluation, ensuring strict independence. Reported test metrics therefore reflect true extrapolation to novel protein families rather than memorization.

Honest reporting

No synthetic or simulated data were used at any stage. The cross-validation-to-test performance gap is reported explicitly. Two S669 numbers are provided — the frozen training-run value and a live re-computation from the deployed model — which differ slightly (≤ 0.03 in correlation) due to ongoing feature-extraction refinements; both are disclosed.

Design tool

An interactive web application accepts any protein sequence or preset PETase/cutinase, runs latent-space optimization to propose stabilizing variants, ranks candidates by combined stability/activity fitness, and visualizes mutations on experimentally accurate 3-D structures with per-residue confidence coloring and per-mutation $\Delta\Delta G$ annotations.

Abstract sections conform to BMES 2026 word limits (Introduction ≤ 250 ; Materials & Methods ≤ 250 ; Results/Conclusions/Discussion ≤ 350). For portal submission, upload figures individually as .png; this PDF is the consolidated reference document. · Generated from *model_meta.json* and live S669 predictions (deployed v46 ensemble).